

## Adsorption of Congo red and tetracycline onto water treatment sludge biochar: characterisation, kinetic, equilibrium and thermodynamic study

Liwenze He<sup>a</sup>, Yu Chen<sup>b,\*</sup>, Yanjun Li<sup>b</sup>, Fei Sun<sup>b</sup>, Yuting Zhao<sup>b</sup> and Shunsheng Yang<sup>a</sup>

<sup>a</sup> School of Civil Engineering, Southwest Jiaotong University, Chengdu 610031, People's Republic of China

<sup>b</sup> Faculty of Geosciences and Environmental Engineering, Southwest Jiaotong University, Chengdu 610031, People's Republic of China

\*Corresponding author. E-mail: chenyu1123@swjtu.edu.cn

### ABSTRACT

In this study, readily available inexpensive water treatment sludge (WTS) was used to prepare adsorbent for the removal of Congo red (CR) and tetracycline (TC) from aqueous solutions. The structural characteristics and adsorption properties of WTS biochar were characterised via scanning electron microscope, energy dispersive X-ray spectroscopy, Brunauer-Emmett-Teller and Fourier Transform infrared spectroscopy. In batch experiments, the adsorption factors, kinetics, isothermal curves and thermodynamics of the adsorption properties were investigated. The optimum preparation condition of WTS biochar was 400 °C for 4 h under O<sub>2</sub>-limited pyrolysis, which exhibited increased specific surface area and pore structures. The best adsorption was observed when the pH of the CR and TC solutions was 7 and 4, respectively. The adsorption process followed the pseudo-second-order model, indicating that the main control step was the chemical adsorption process. Isotherm data were best described by the Langmuir model, and the maximum adsorption capacities for CR and TC were 116.4 and 58.5 mg·g<sup>-1</sup>, respectively. Thermodynamic parameters revealed that the adsorption process was spontaneous and endothermic. According to the analysis, the adsorption mechanism of CR could be attributed to electrostatic attraction,  $\pi$ - $\pi$  conjugation and hydrogen bonding, whereas that of TC was potentially associated with cation exchange, complex precipitation,  $\pi$ - $\pi$  conjugation and hydrogen bonding.

**Key words:** adsorption, Congo Red, kinetics, tetracycline, thermodynamics, water treatment sludge

### HIGHLIGHTS

- Synthetic dyes and antibiotics have been widely used and are difficult to biodegrade.
- In this study, water treatment sludge was prepared, and its adsorption properties of the dye and antibiotics were investigated.
- It is a promising adsorbent with stable and efficient adsorption capacity, low cost, environmental protection and safety.

## 1. INTRODUCTION

Synthetic dyes have been widely used in textile, leather manufacturing and papermaking and other fields. In addition to bringing economic value, dyes have the characteristics of complex chemical structure, high stability and high toxicity. Moreover, most dyes are difficult to biodegrade, and their metabolic intermediates possess mutagenesis and carcinogenesis potential (Sun *et al.* 2011). Wastewater containing dyes hardly meets emission requirements after biological treatment, which causes acute and chronic problems to the ecological system and become one of the most hard-to-treat kinds of wastewater (Guo *et al.* 2012). Azo dye Congo red (CR), which has a high loss rate in the process of production and use, is a representative pollutant in printing and dyeing wastewater (Hua *et al.* 2018; Patel 2018).

Recently, as a type of emerging trace pollutant, antibiotics have been frequently detected in natural water, which has attracted widespread concern (Sui *et al.* 2012). Tetracycline (TC) has become one of the most widely used antibiotics globally because of its remarkable efficacy and low price. It is difficult for TC to be completely metabolised by human and animal bodies; therefore, it is discharged into water in large quantities (Martins *et al.* 2015). Existing wastewater treatment systems cannot effectively remove TC (Miao *et al.* 2004). It is often detected in surface water, soil and vegetables, posing a severe threat to human health (Schmitt *et al.* 2006; Akiyama & Savin 2010). At present, the primary methods for removing dyes and antibiotics from aqueous solutions include adsorption (Liu *et al.* 2010; Peng *et al.* 2012; Markandeya *et al.* 2021), advanced oxidation (Santos *et al.* 2015; Yan & Wang 2016), biodegradation and membrane separation (Amorim *et al.*

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2014; Hussein & Scholz 2018). Among these technologies, adsorption is considered an effective and inexpensive method to remove pollutants from wastewater without producing more toxic or difficult-to-degrade pollutants than was begun with. In addition, factors, such as low cost, easy availability, high yield, adsorption performance and availability of adsorbents, must be considered. Hence, it is essential to seek an optimal adsorbent that meets the above requirements.

Water treatment sludge (WTS) is a by-product of coagulation–sedimentation–filtration and other treatment processes in water supply plants. The main components are primarily comprise amorphous masses of iron and aluminium hydroxides. It also contains sediments carried in raw water, humic substances and dissolved colloids (Babatunde & Zhao 2007; Abo-El-Enein *et al.* 2017). Owing to urbanisation and population growth, the output of WTS is increasing yearly, and the conventional landfill means are unsuitable. Therefore, resource utilisation will become an inevitable trend. WTS is rich in iron and aluminium, has a large specific surface area, a big internal pore size and a high yield and is inexpensive and simple to obtain, which are significant advantages of an adsorbent. Because of its efficient adsorption performance to water pollutants, new applications of WTS in water purification are constantly being tried. In recent years, reuse of WTS as low-cost adsorbent for phosphorus immobilisation represents a large number of investigations (Ahmad *et al.* 2016; Hidalgo *et al.* 2017). It has been reported that oxytetracycline is adsorbed by WTS under the optimal condition of  $20 \text{ g} \cdot \text{L}^{-1}$ , and more than 95% oxytetracycline is adsorbed within 2 h (Punamiya *et al.* 2013). Makris carried out the adsorption and desorption of  $\text{ClO}_4^-$  by WTS (Makris *et al.* 2006). To expand the scope of reuse, WTS has been tested for adsorption of a number of heavy metals, which include Cd, Cu, Pb, Hg (Shen *et al.* 2019). Nevertheless, there are few reports about using WTS as the source material for dye and antibiotic adsorption. The adsorption properties and mechanism by which WTS adsorbs different types of contaminants have not been clearly elucidated.

In this study, WTS was prepared by an  $\text{O}_2$ -limited heated process, and its adsorption properties for dye and antibiotics were investigated. CR and TC were selected as typical anion dye and antibiotic, respectively. The effects of pH, biochar dosage, initial concentration, temperature and adsorption time on the adsorption effect were investigated, and the physicochemical properties of WTS biochar before and after adsorption of CR and TC were characterised. On this basis, the adsorption mechanism was analysed by the adsorption kinetic, isotherm model and thermodynamic model to provide a theoretical basis and a new approach for the resource utilisation of WTS.

## 2. MATERIAL & METHODS

### 2.1. Raw materials and biochar preparation

Congo red and tetracycline were purchased from Tianjin Kermel Chemical Reagent Co., Ltd. The  $\text{KNO}_3$ , NaOH, HCl and other reagents used in the experiment were all analytical reagents and purchased from Sinopharm Chemical Reagent Co., Ltd. Ultrapure water was used in all experiments.

Instruments: muffle furnace (SX2-8-10, Shanghai Institute of Light); acidity meter (PH-3C, Shanghai ray magnetic); UV–vis spectrophotometer (UV – 1800PC, Japan HITACHI); electronic analytical balance (JT2003, Shanghai lichen); gas bath thermostat oscillator (THZ-82, Jintan Yineng experimental instrument factory); Fourier Infrared Spectrometer (Nicolet 670, Thermo Nicolet Corporation); field emission scanning electron microscope (PV77-47170 ME, USA EDAX); specific surface area analyser (ASAP 2460, America Micromeritics).

Raw water treatment sludge (RS) was obtained from a drinking water treatment plant in Chengdu, which used PAC and PAM as flocculants. After being dried in an oven at  $105^\circ\text{C}$  for 24 h, the sludge was ground with a mortar and sifted through a 60-mesh sieve to remove large particles.

An appropriate amount of RS was placed into a crucible. The crucible was then put in a muffle furnace and heated under the  $\text{O}_2$ -limited condition at each of 300, 400, 500, 600 and  $700^\circ\text{C}$ , with the heating rate set at  $10^\circ\text{C} \cdot \text{min}^{-1}$  and holding times were 1, 2, 4 and 6 h. After pyrolysis finished, the biochars were milled with a mortar and pestle to penetrate a 100-mesh sieve and collected to keep in a desiccator for characterisation and sorption experiment. According to different temperature and time conditions, the biochars obtained were named  $\text{WS}_{300-2}$ ,  $\text{WS}_{400-2}$ ,  $\text{WS}_{500-2}$ ,  $\text{WS}_{600-2}$ ,  $\text{WS}_{700-2}$ ,  $\text{WS}_{400-1}$ ,  $\text{WS}_{400-4}$  and  $\text{WS}_{400-6}$ .

### 2.2. Biochar characterisation

The scanning electron microscopy-energy dispersive X-ray spectroscopy (SEM-EDS) provided morphological information and the main chemical composition of the biochar surface. Fourier transform infrared spectroscopy (FTIR) analysis was employed to verify the presence of surface functional groups on the biochar and raw sludge. The spectra ranged from

4,000 to 400  $\text{cm}^{-1}$ . The Brunauer–Emmett–Teller (BET) method was used to determine the specific surface area, total pore volume and average pore size of WTS samples.

The zero point charge ( $\text{pH}_{\text{ZPC}}$ ) of the adsorbent is a key factor affecting the adsorption capacity. In general,  $\text{pH} > \text{pH}_{\text{ZPC}}$  is favourable for the adsorption of cations, whereas  $\text{pH} < \text{pH}_{\text{ZPC}}$  is favourable for the adsorption of anions. The zero point charge ( $\text{pH}_{\text{ZPC}}$ ) of  $\text{WS}_{400-4}$  was determined by the following method (Liu & Zhang 2011; Liu *et al.* 2011):

- (1) A 50-mL sodium chloride ( $\text{NaCl}$ , 0.01 M) solution was placed into a 100-mL conical flask with a stopper. Afterwards, 0.01 M  $\text{HCl}$  and  $\text{NaOH}$  were used to adjust the successive initial solution pH values from 2 to 12, and 0.25 g of the  $\text{WS}_{400-4}$  sample was added to the conical flask.
- (2) The conical flask was filled with  $\text{N}_2$  to eliminate the effect of carbon dioxide on the pH change, and then the mixture was placed in an oscillator at 303 K for 48 h reaction.
- (3) The difference between the final pH and the initial pH, denoted as  $\Delta\text{pH}$ , was plotted against the initial pH. The solution pH at which the curve crossed the line of  $\Delta\text{pH} = \text{pH}(\text{final}) - \text{pH}(\text{initial}) = 0$  was taken as the  $\text{pH}_{\text{ZPC}}$  of the sample (Zhu *et al.* 2014).

### 2.3. Adsorption experiments

0.2 g of RS,  $\text{WS}_{300-2}$ ,  $\text{WS}_{400-2}$ ,  $\text{WS}_{500-2}$ ,  $\text{WS}_{600-2}$ ,  $\text{WS}_{700-2}$ ,  $\text{WS}_{400-1}$ ,  $\text{WS}_{400-4}$  and  $\text{WS}_{400-6}$  were weighed into a 250 mL conical flask filled with 100 mL of 100  $\text{mg}\cdot\text{L}^{-1}$  CR and TC solutions, respectively. The initial pH was unadjusted. Next, all adsorption experiments were performed via a batch technique in a thermostat oscillator at a shaking rate of 150 rpm and 30 °C for 12 h to adsorption equilibrium. After the adsorption, the mixture was filtered by a 0.45- $\mu\text{m}$  inorganic filter membrane and the remaining concentration of CR and TC were determined using a UV-vis spectrophotometer. The equilibrium adsorption capacity ( $q_e$ ,  $\text{mg}/\text{g}$ ) and removal rate ( $R$ , %) were calculated using Equations (1)–(3) respectively. All batch experiments were performed in triplicate, and the average data were presented.

$$q_e = \frac{(C_0 - C_e)V}{M} \quad (1)$$

$$q_t = \frac{(C_0 - C_t)V}{M} \quad (2)$$

$$R = \frac{C_0 - C_e}{C_0} \times 100\% \quad (3)$$

where  $C_0$  ( $\text{mg}\cdot\text{L}^{-1}$ ) is the initial concentration of CR and TC;  $C_e$  ( $\text{mg}\cdot\text{L}^{-1}$ ) is the equilibrium concentration of CR and TC;  $C_t$  is the concentration at contact time of  $t$  (min);  $q_e$  ( $\text{mg}\cdot\text{g}^{-1}$ ) means that the amount of CR and TC adsorption at equilibrium;  $q_t$  ( $\text{mg}\cdot\text{g}^{-1}$ ) means the amount of CR and TC adsorbed by biochar at contact time of  $t$  (min);  $M$  (g) is biochar mass;  $V$  (L) means the volume solution;  $R(\%)$  is the removal ratio.

Effect of different factors on adsorption of CR and TC: (1) Initial pH effect: 0.1 M  $\text{HCl}$  or  $\text{NaOH}$  was added to the 100 mL of 100  $\text{mg}\cdot\text{L}^{-1}$  solutions to adjust the pH to 3–9, then 0.2 g of  $\text{WS}_{400-4}$  was weighed into each solution of different pH. In subsequent experiments, the pH of the CR and TC solutions was adjusted to 7 and 4, respectively. (2) Effect of biochar dosage: different dosages of  $\text{WS}_{400-4}$  were added to the 100  $\text{mg}\cdot\text{L}^{-1}$  CR and TC solutions with liquid–solid ratios ranging from 0.2 to 4  $\text{g}\cdot\text{L}^{-1}$ . (3) Effect of initial adsorption concentration: 0.2 g of  $\text{WS}_{400-4}$  was added to the CR and TC solutions at concentrations of 60–200  $\text{mg}\cdot\text{L}^{-1}$ . (4) Effect of temperature: 0.2 g of  $\text{WS}_{400-4}$  was added to solutions which were placed in a thermostatic oscillator and oscillated for 12 h at 10, 20, 30, 40 and 50 °C to adsorption equilibrium.

### 2.4. Adsorption kinetic analysis

$\text{WS}_{400-4}$  (2  $\text{g}\cdot\text{L}^{-1}$ ) was added to different initial concentrations of CR (100 and 200  $\text{mg}\cdot\text{L}^{-1}$ ) and TC (120 and 180  $\text{mg}\cdot\text{L}^{-1}$ ); then, the adsorption reaction was performed at 30 °C and 150 rpm. The samples were measured at 0, 5, 10, 15, 20, 30, 40, 60, 120, 180, 240, 360, 480, 600 and 720 min. To understand the adsorption process, three adsorption kinetic models were analysed.

Pseudo-first-order kinetic model

$$\log (q_e - q_t) = \log q_e - k_1 t \quad (4)$$

Pseudo-second-order kinetic model

$$\frac{t}{q_t} = \frac{1}{k_2 q_e^2} + \frac{t}{q_e} \quad (5)$$

Particle diffusion model

$$q_t = k_3 t^{0.5} + A \quad (6)$$

where  $q_t$  ( $\text{mg}\cdot\text{g}^{-1}$ ) and  $q_e$  ( $\text{mg}\cdot\text{g}^{-1}$ ) are the amounts of CR adsorbed by biochar at contact time  $t$  and equilibrium, respectively;  $k_1$  ( $\text{min}^{-1}$ ),  $k_2$  ( $\text{g}\cdot(\text{mg}\cdot\text{min})^{-1}$ ), and  $k_3$  ( $\text{mg}\cdot(\text{g}\cdot\text{min}^{0.5})^{-1}$ ) are the rate constants of pseudo-first-order, pseudo-second-order, and intra-particle diffusion models, respectively;  $t$  (min) is the adsorption time;  $A$  is a constant related to the thickness of the boundary layer thickness.

## 2.5. Adsorption isotherms and thermodynamic models

According to the parameters observed from the experiments described above, 0.2 g of WS<sub>400-4</sub> was added to different initial concentrations of CR (60–200  $\text{mg}\cdot\text{L}^{-1}$ ) and TC (20–200  $\text{mg}\cdot\text{L}^{-1}$ ). The experiment results were analysed with Langmuir and Freundlich models (Equations (7) and (8), respectively) to evaluate the adsorption (Li *et al.* 2005; Ünlü & Ersoz 2006). Thermodynamic parameters, such as  $\Delta G^\theta$ ,  $\Delta H^\theta$  and  $\Delta S^\theta$ , for CR and TC adsorption, were calculated using Equations (9) and (10).

$$\frac{1}{q_e} = \frac{1}{q_m} + \frac{1}{q_m K_L C_e} \quad (7)$$

$$\log q_e = \log K_F + \frac{1}{n} \log C_e \quad (8)$$

where  $K_L$  ( $\text{L}\cdot\text{mg}^{-1}$ ) and  $K_F$  ( $\text{mg}\cdot\text{g}^{-1}$ ) are the constants in Langmuir and Freundlich models, respectively;  $q_m$  ( $\text{mg}\cdot\text{g}^{-1}$ ) is the theoretical maximum of adsorption amount;  $C_e$  ( $\text{mg}\cdot\text{L}^{-1}$ ) is the equilibrium concentration;  $1/n$  is the heterogeneity factor indicating how favourable the adsorption process is.

$$\Delta G^\theta = -RT \ln K_d \quad (9)$$

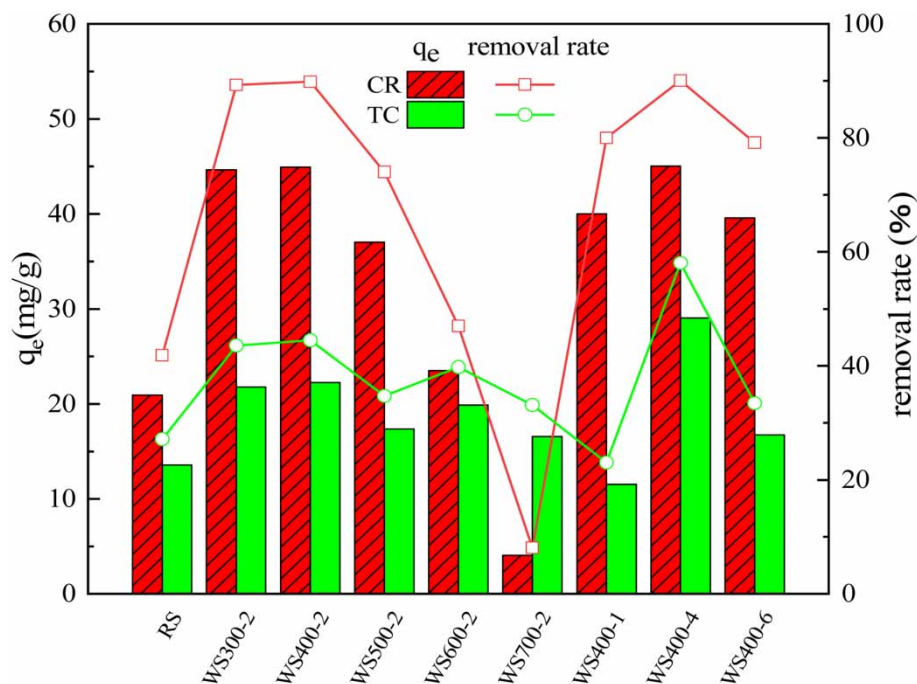
$$\Delta G^\theta = \Delta H^\theta - T \Delta S^\theta \quad (10)$$

Plot  $\ln q_e/C_e$  with respect to  $q_e$ , and then extend to the intercept to obtain the value of  $\ln K_d$ ; where  $R$  ( $8.314 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$ ) is the ideal gas constant,  $T(\text{K})$  is Kelvin temperature,  $\Delta G^\theta$  ( $\text{kJ}\cdot\text{mol}^{-1}$ ) is the change in the Gibbs free energy,  $\Delta H^\theta$  ( $\text{kJ}\cdot\text{mol}^{-1}$ ) is the change in enthalpy,  $\Delta S^\theta$  ( $\text{kJ}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$ ) is the change in entropy.

## 3. RESULTS AND DISCUSSION

### 3.1. Determination of optimal preparation conditions

The adsorption capacity and removal rate of CR and TC by biochar prepared under different conditions are shown in Figure 1. With an increase in preparation temperature from 300 °C to 400 °C, the removal rates of CR and TC of biochar increased slightly. When the temperature increased from 400 °C to 700 °C, the adsorption capacity of CR decreased to  $4.05 \text{ mg}\cdot\text{g}^{-1}$  and the removal rate was only 8.10%, whereas the adsorption capacity of TC decreases to  $16.56 \text{ mg}\cdot\text{g}^{-1}$  and the removal rate was 33.14%. It may be that high temperature leads to structure collapse, pore blockage, specific surface area reduction, and adsorption capacity decline (Liu *et al.* 2007). The fixed reaction temperature was 400 °C, and the removal of pollutants increased first and then decreased with the extension of pyrolysis time. When the reaction temperature was 400 °C and the



**Figure 1** | Effect of different biochars on CR and TC adsorption.

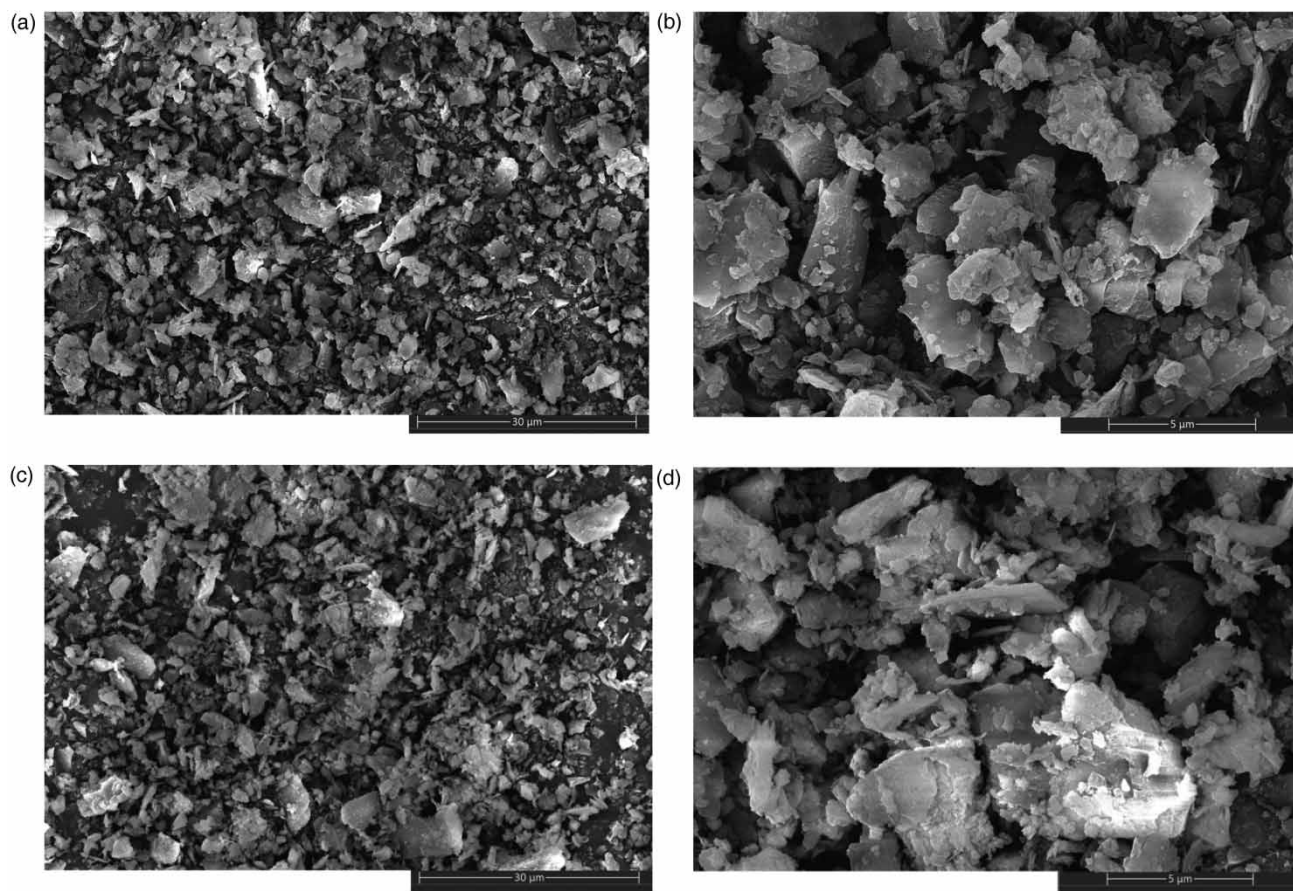
reaction time was 4 h, the biochar had an optimal adsorption effect on CR and TC. Therefore, 400 °C and 4 h were selected as the best reaction conditions for the preparation of biochar.

### 3.2. Characterisation of adsorbents

Figure 2 presents the SEM micrographs of RS and WS<sub>400-4</sub>. The surface morphology of RS is a closely connected lamellar structure. After heating treatment, the surface of the lamellar structure became rough. This may be due to the decomposition of organic matter in the WTS by heating, which changed the morphology, increased the pore size and loosened the structure (Nguyen *et al.* 2019; Wu *et al.* 2020). Moreover, the stable Si-Al lattice structure was damaged, the surface was rougher and larger pores were formed inside, which was also reflected in the element composition in Figure 3. Compared with RS, the carbon content in WS<sub>400-4</sub> decreased significantly, the effective pore size increased, and the contents of inorganic matter, Fe, Al and Ca increased. There was no significant change in the elemental composition of RS and WS<sub>400-4</sub>, and the main elemental composition was C, O, Na, Al, Si, Ca, Fe, etc., accounting for more than 85% of the total proportion, which agrees with the composition of WTS (Yang *et al.* 2015). The pore structure analysis results of biochar are shown in Table 1. Under the optimum preparation conditions (temperature 400 °C and reaction time 4 h), the BET surface area increased slightly from 29.39 m<sup>2</sup>/g to 34.22 m<sup>2</sup>/g. The increase in the specific surface area increased the active site for CR and TC adsorption. The total pore volume increased from 0.091 cm<sup>3</sup>/g to 0.127 cm<sup>3</sup>/g, and the average pore diameter increased by 1.35 times from 7.77 nm to 10.5 nm. It shows that thermal modification of WTS can effectively improve the pore diffusion adsorption capacity. Zero charge point (pH<sub>zpc</sub>) refers to the pH value of the solution when the net charge on the surface of carbon material is zero. According to pH<sub>zpc</sub>, we can not only provide a theoretical basis for the adsorption of carbon material, but also predict the adsorption capacity of carbon material under different pH values. Figure 4 depicts the pH<sub>zpc</sub> of the biochar of WS<sub>400-4</sub>. When the pH value of the solution is 8.9, the numbers of positive and negative charges of biochar are equal. When the pH value of the solution is less than pH<sub>zpc</sub>, the surface of biochar is positively charged and can adsorb anions in the solution. Therefore, the higher the pH<sub>zpc</sub> value is, the more likely the adsorbent is to be positively charged in a wider pH range, which is more conducive to adsorption through electrostatic action.

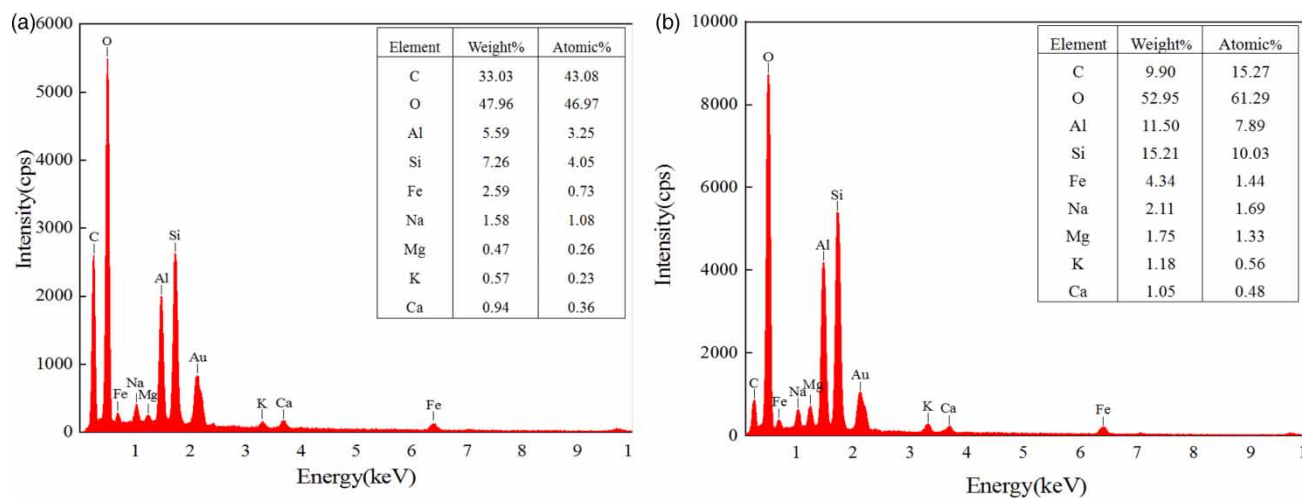
### 3.3. Effect of solution pH

The pH value of the solution further affects the adsorption process by affecting the adsorbent and the adsorb surface charge and form. The effects of different initial solution pH values on the adsorption of CR and TC are shown in Figure 5. With the



**Figure 2** | SEM figure (a) (b) RS(raw water treatment sludge); (c) (d) WS<sub>400-4</sub>.

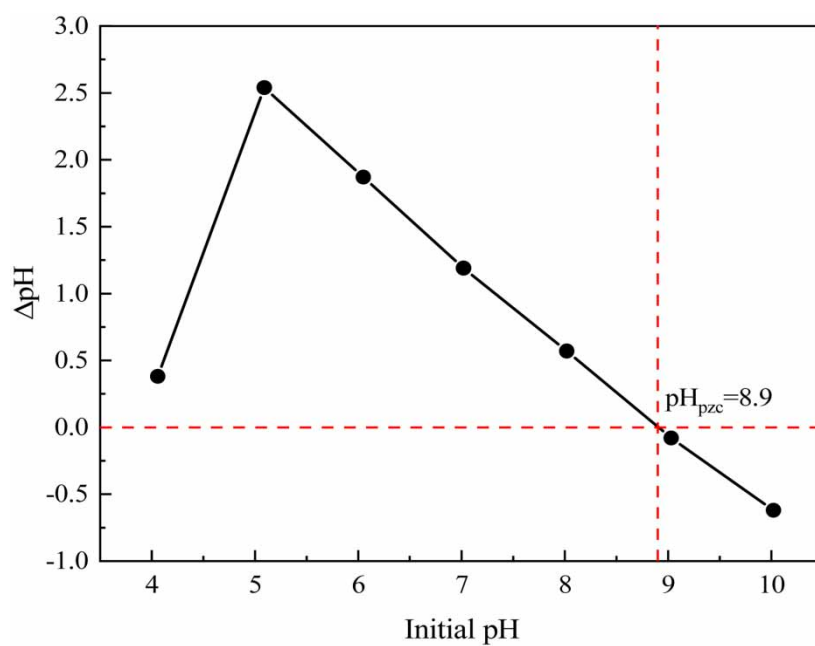
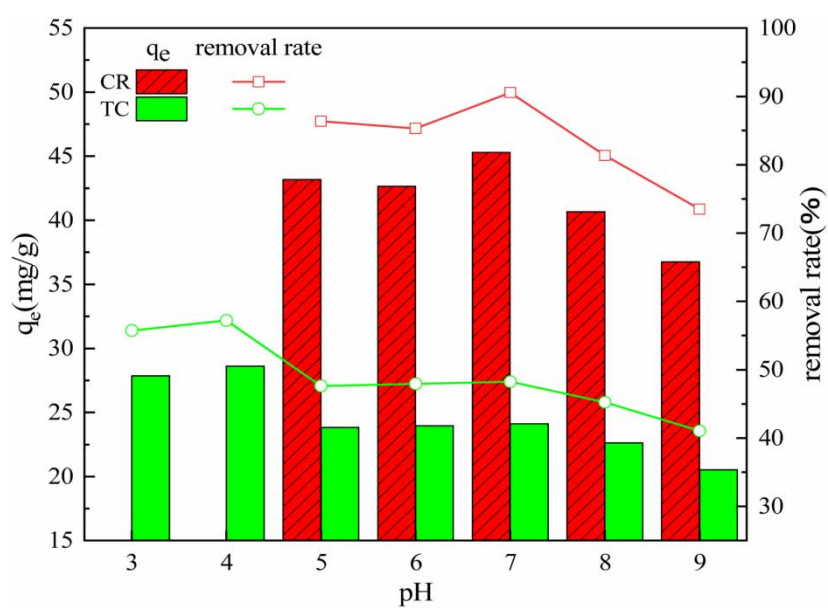
increase of pH value from 5 to 7, the CR removal rate of WS<sub>400-4</sub> increased from 83.36% to 90.57%. Under acidic conditions, CR ionises and becomes negative, and the surface of WS<sub>400-4</sub> is positively charged at this time, due to the increased adsorption of CR by electrostatic attraction. As the solution pH value increases to 9 and tends to be alkaline, the solution pH value is greater than the  $pH_{zpc}$  of WS<sub>400-4</sub>, making the WS<sub>400-4</sub> surface negatively charged, thereby repelling CR and significantly reducing its adsorption of CR.



**Figure 3** | EDS figure (a) RS; (b) WS<sub>400-4</sub>.

**Table 1** | BET analysis of biochars

| Biochars            | Surface area/(m <sup>2</sup> ·g <sup>-1</sup> ) | Total pore volume/(cm <sup>3</sup> ·g <sup>-1</sup> ) | Average pore diameter/(nm) |
|---------------------|-------------------------------------------------|-------------------------------------------------------|----------------------------|
| RS                  | 29.39                                           | 0.091                                                 | 7.77                       |
| WS <sub>400-4</sub> | 34.22                                           | 0.127                                                 | 10.5                       |

**Figure 4** | The pH<sub>zpc</sub> determination of WS<sub>400-4</sub>.**Figure 5** | Effect of pH on adsorption.

TC has three dissociation equilibrium constants,  $\text{pK}_{\text{a}1} = 3.3$ ,  $\text{pK}_{\text{a}2} = 7.68$ ,  $\text{pK}_{\text{a}3} = 9.69$  (Cao *et al.* 2018). The morphology and electrification of TC differ for different pH values. The main ionic species of TC were  $\text{TC}^+$  at  $\text{pH} < 3.3$ ,  $\text{TC}^\pm$  at  $3.3 < \text{pH} < 7.68$ ,  $\text{TC}^-$  at  $7.68 < \text{pH} < 9.69$  and  $\text{TC}^{2-}$  at  $\text{pH} > 9.69$  (Yan *et al.* 2020). When the pH of the adsorption system is 3–4,  $\text{WS}_{400-4}$  is positively charged on the surface, TC exists in the form of cation at this time, and there is electrostatic repulsion between the two. However, in this study, the removal rate was relatively high and showed an increasing trend, indicating that there may be other forces such as coordination complexation or ion exchange affecting the adsorption process.  $\text{WS}_{400-4}$  has high ash content and is rich in mineral cations such as  $\text{K}^+$ ,  $\text{Na}^+$ ,  $\text{Fe}^{3+}$ ,  $\text{Mg}^{2+}$  and  $\text{Ca}^{2+}$ . It has been reported that cationic TC can be combined with the surface coordination sites of biochar and cation exchange reaction occurs with coordinated cations (Saravanan *et al.* 2016; Qu *et al.* 2019), resulting in a high removal rate of TC. When the pH value is 5–8, TC mainly exists in the form of cations and neutral molecules, and the electrostatic repulsion between them inhibits the adsorption of  $\text{WS}_{400-4}$  on TC. It has been reported that iron oxide has strong coordination and a complex effect on TC (Zhu *et al.* 2014). The removal rate of TC decreased significantly with pH increase, probably because, under acidic conditions with low pH value, cations such as iron, calcium and magnesium in biochar are dissolved and the coordination reaction between iron ion and TC occurs, resulting in adsorption or precipitation after polymerisation with TC (Tongaree *et al.* 1999; Arias *et al.* 2007; Liang *et al.* 2019).

As the initial pH of the solution increases from 5 to 7, the ionisation property of TC gradually changes from electropositive to electroneutral, the electrostatic repulsion with adsorbent decreases gradually, and the adsorption capacity increases slightly. However, the adsorption capacity decreases significantly when the pH is higher than 4, indicating that cation exchange and complex precipitation contribute more to adsorption than electrostatic action. At  $\text{pH} = 9$ ,  $\text{WS}_{400-4}$  is negatively charged on the surface and repels TC in the form of anionic ions, thereby reducing the removal rate considerably. In summary, the optimal pH conditions for CR and TC adsorption are 7 and 4, respectively. We infer that the adsorption process of CR was electrostatic attraction, whereas the adsorption process of TC was cation exchange, complex precipitation.

### 3.4. Effect of adsorbent dosage

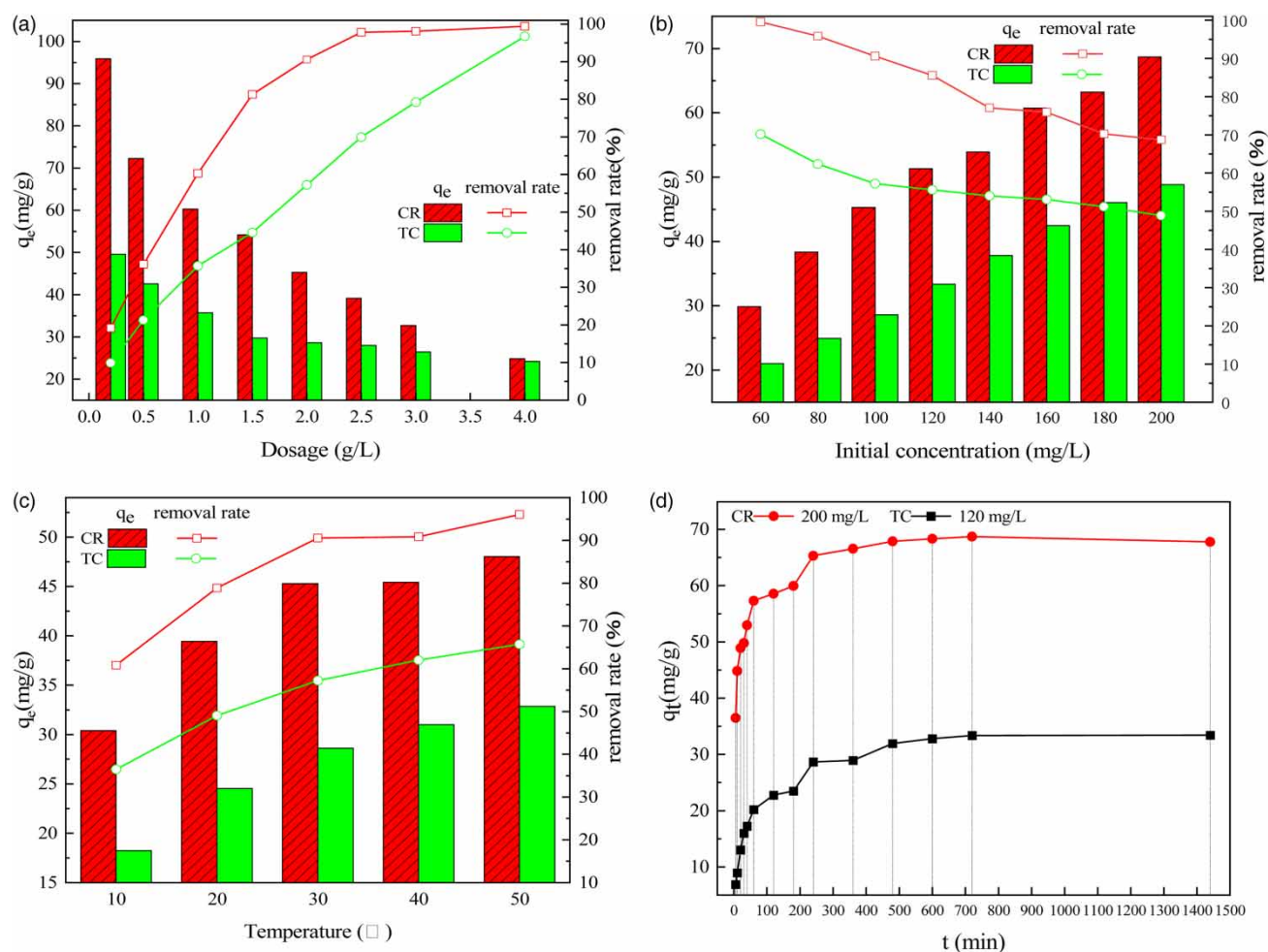
According to Figure 6(a), with an increase in adsorbent dosage, more active adsorption sites were provided, the removal rate of CR increased from 19.18% to 99.39%, and TC rose from 9.92% to 96.72%. In addition, because the total concentrations of pollutants were the same, the increase of adsorbent will reduce the driving force of each adsorption activity point (Wang *et al.* 2014), thereby making the adsorption amount present a downward trend. The adsorption capacity of CR decreased from  $95.90 \text{ mg}\cdot\text{g}^{-1}$  to  $24.85 \text{ mg}\cdot\text{g}^{-1}$ , and that of TC decreased from  $49.58 \text{ mg}\cdot\text{g}^{-1}$  to  $24.18 \text{ mg}\cdot\text{g}^{-1}$ . Considering the economy and removal efficiency, the dosage is  $2 \text{ g}\cdot\text{L}^{-1}$ , and the adsorption capacity of CR and TC at adsorption equilibrium is 45.28 and  $28.61 \text{ mg}\cdot\text{g}^{-1}$ , respectively.

### 3.5. Effect of initial concentration and temperature

The effects of initial concentration and temperature are shown in Figure 6(b) and 6(c). The equilibrium adsorption capacity of CR and TC increased with an increase in the initial concentration of CR and TC, and the increase in ambient temperature also promoted the adsorption of CR and TC. When the temperature increased from  $10^\circ\text{C}$  to  $50^\circ\text{C}$ , the equilibrium adsorption capacity of CR increased from  $30.40 \text{ mg}\cdot\text{g}^{-1}$  to  $48.04 \text{ mg}\cdot\text{g}^{-1}$ , and the removal rate increased from 60.81% to 96.08%. The constant adsorption capacity of TC increased from  $18.24 \text{ mg}\cdot\text{g}^{-1}$  to  $32.86 \text{ mg}\cdot\text{g}^{-1}$ . Based on these results, when the temperature increased, the diffusion rate of CR and TC increased, thereby increasing the amounts of CR and TC adsorbed and removal rates.

### 3.6. Effect of contact time

The effects of different contact times on the adsorption process are shown in Figure 6(d). The adsorption of CR and TC by  $\text{WS}_{400-4}$  shows the characteristics of fast adsorption and slow equilibrium. In the first 4 h, the adsorption capacity increases rapidly and the growth rate tends to be slow from 4 to 12 h. The dye and drug molecules are first rapidly adsorbed on the active sites on the adsorbent surface, and then the system advances to the equilibrium state of adsorption and desorption. Therefore, 12 h was selected as the adsorption equilibrium time point.

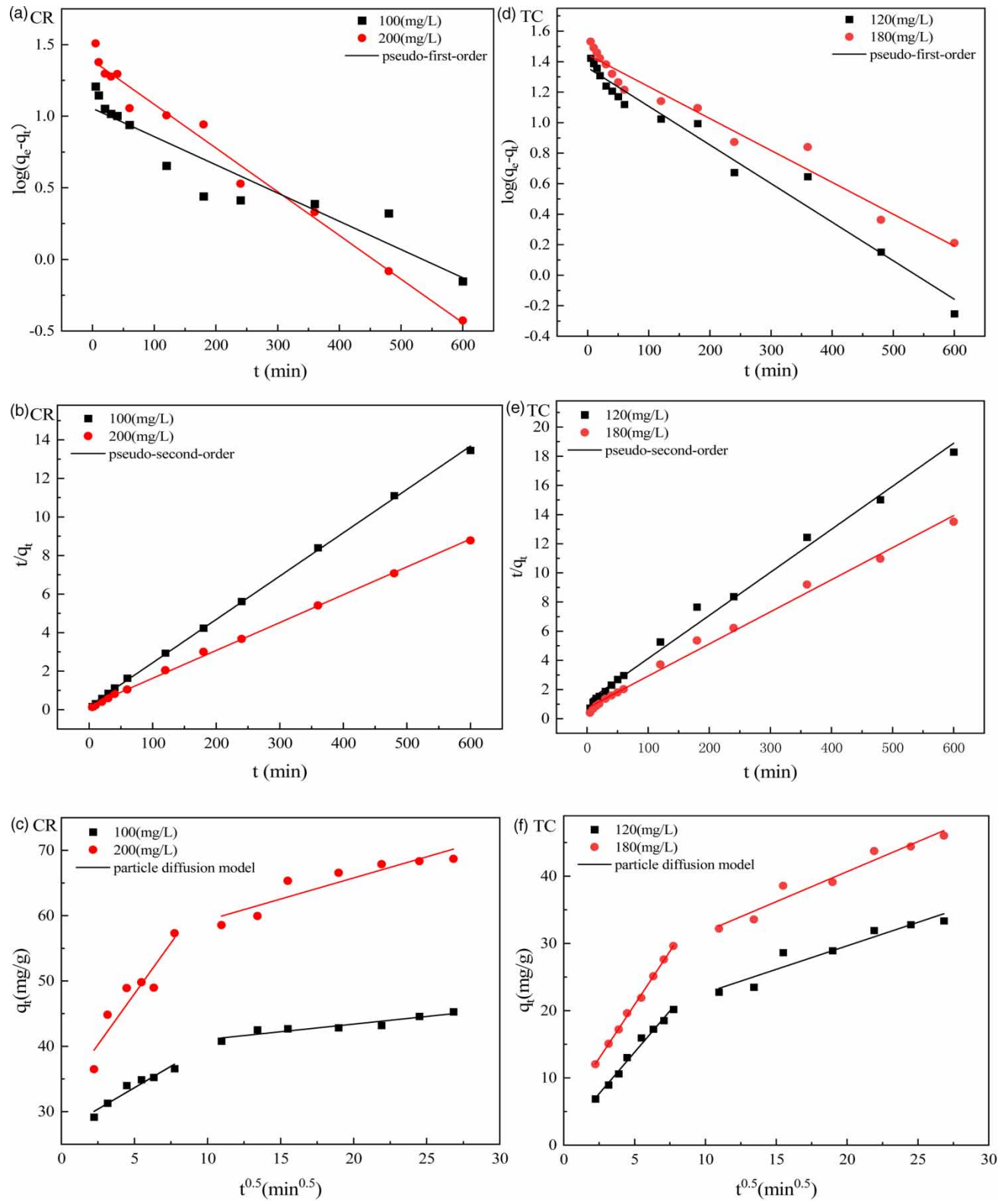


**Figure 6** | Effects of different variables on adsorption efficiency (a) biochar dosage, (b) CR and TC concentration, (c) temperature, and (d) contact time.

### 3.7. Adsorption kinetics

The adsorption kinetic curves of CR and TC onto WS<sub>400-4</sub> are presented in Figure 7, and the parameters are shown in Table 2. From the fitting results that the pseudo-second-order model ( $R^2 > 0.99$ ) was superior to the pseudo-first-order model, the theoretical equilibrium adsorption quantity  $q_{e,cal}$  calculated was close to the experimental equilibrium adsorption quantity  $q_{e,exp}$ , and the relative error was less than 3.89%. Therefore, the pseudo-second-order model can more accurately describe the adsorption process of CR and TC by WS<sub>400-4</sub>. The pseudo-second-order equation included adsorption processes such as liquid film diffusion, surface adsorption and intra-particle diffusion, and electron sharing or transferring (Fan *et al.* 2016). The results showed that the main control step of CR and TC adsorption by WS<sub>400-4</sub> was the chemical adsorption process, and there was electron transfer between adsorbent and pollutant (Vadivelan & Kumar 2005; Simonin 2016). It can be speculated that the adsorption of CR and TC by WS<sub>400-4</sub> was not a single mechanism of action.

To further explore the steps of adsorption speed control, the data of CR and TC were fitted by the intra-particle diffusion model. The graph was divided into two parts, indicating that the whole diffusion process was divided into two steps. The first part was the liquid film diffusion process, and  $k_{3,1}$  was larger, indicating that the boundary diffusion process was faster. The second stage was the internal diffusion process, and  $k_{3,2}$  was smaller, indicating that the internal diffusion process was the rate-limiting step. According to this model, if the line does not pass the origin, it means that intra-particle diffusion is not the only controlling step, and other processes control the reaction rate. According to the relevant results, the line never crossed the origin, indicating that the adsorption rate of CR and TC may be controlled by liquid film diffusion, intra-particle diffusion and surface adsorption (Wei *et al.* 2019).



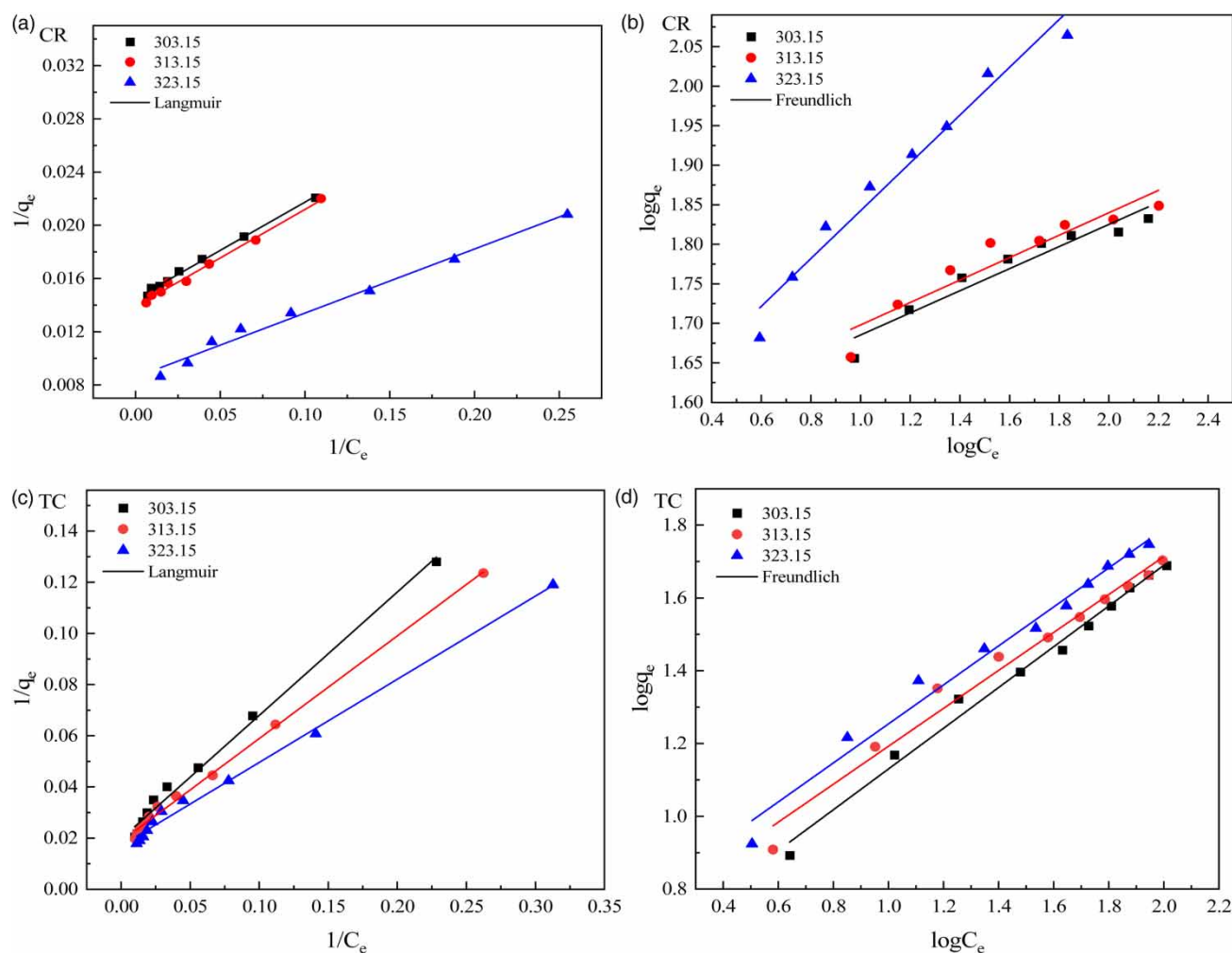
**Figure 7** | Adsorption kinetic curves of CR and TC onto biochars.

**Table 2** | Adsorption kinetics parameters

|    |          | Pseudo-first-order kinetic model |             |        | Pseudo-second-order kinetics model |             |        | Intra-particle diffusion model |           |        |         |           |        |
|----|----------|----------------------------------|-------------|--------|------------------------------------|-------------|--------|--------------------------------|-----------|--------|---------|-----------|--------|
|    |          | $k_1$                            | $q_{e,cal}$ | $R^2$  | $k_2$                              | $q_{e,cal}$ | $R^2$  | $A_1$                          | $k_{3,1}$ | $R^2$  | $A_2$   | $k_{3,2}$ | $R^2$  |
| CR | 100 mg/L | 0.0019                           | 11.339      | 0.8804 | 0.0027                             | 44.504      | 0.9999 | 27.089                         | 1.3128    | 0.9642 | 38.701  | 0.2347    | 0.9418 |
|    | 200 mg/L | 0.0031                           | 24.422      | 0.9813 | 0.0011                             | 69.252      | 0.9987 | 32.434                         | 3.1181    | 0.9289 | 52.845  | 0.6466    | 0.9208 |
| TC | 120 mg/L | 0.0025                           | 22.925      | 0.9721 | 0.0007                             | 33.898      | 0.9926 | 1.4946                         | 2.4654    | 0.9886 | 15.6531 | 0.6996    | 0.9155 |
|    | 180 mg/L | 0.0021                           | 27.881      | 0.9674 | 0.0006                             | 45.455      | 0.9935 | 4.9613                         | 3.1854    | 0.9983 | 22.8119 | 0.8934    | 0.9525 |

### 3.8. Adsorption isotherms and thermodynamics

The adsorption isotherm is the curve of the relationship between the concentration of adsorbate in the adsorbent and the solution when the adsorption equilibrium is reached at a specific temperature (Zhang *et al.* 2018). Adsorption temperatures were set at 303.15, 313.15 and 323.15 K. Langmuir and Freundlich isothermal model equations were fitted for experimental adsorption data; the results are shown in Figure 8 and Table 3. Both isothermal models can well describe the entire adsorption process at different temperatures. The equilibrium adsorption capacity increased with ambient temperature. The  $R^2$  of the Langmuir model was greater than 0.98 and closer to 1 at different temperatures, whereas the  $R^2$  of the Freundlich

**Figure 8** | Isothermal adsorption curves of CR and TC onto biochars.

**Table 3** | Isotherm equation and thermodynamic parameters

|    | T/K    | Langmuir |        |        | Freundlich |        |        | $\Delta G^\theta$<br>kJ·mol <sup>-1</sup> | $\Delta H^\theta$<br>kJ·mol <sup>-1</sup> | $\Delta S^\theta$<br>J·mol <sup>-1</sup> ·K <sup>-1</sup> |
|----|--------|----------|--------|--------|------------|--------|--------|-------------------------------------------|-------------------------------------------|-----------------------------------------------------------|
|    |        | $q_m$    | $K_L$  | $R^2$  | $1/n$      | $K_F$  | $R^2$  |                                           |                                           |                                                           |
| CR | 303.15 | 69.013   | 0.1998 | 0.9951 | 0.1398     | 35.119 | 0.9147 | -3.945                                    | 37.731                                    | 136.31                                                    |
|    | 313.15 | 71.942   | 0.1905 | 0.9916 | 0.1419     | 35.991 | 0.8756 | -4.177                                    |                                           |                                                           |
|    | 323.15 | 116.410  | 0.1784 | 0.9845 | 0.3023     | 34.687 | 0.9638 | -6.730                                    |                                           |                                                           |
| TC | 303.15 | 50.659   | 0.0409 | 0.9918 | 0.5599     | 3.716  | 0.9899 | -1.851                                    | 20.162                                    | 72.89                                                     |
|    | 313.15 | 53.248   | 0.0468 | 0.9969 | 0.5212     | 4.690  | 0.9789 | -2.844                                    |                                           |                                                           |
|    | 323.15 | 58.582   | 0.0525 | 0.9943 | 0.5359     | 5.223  | 0.9813 | -3.297                                    |                                           |                                                           |

model was between 0.87 and 0.98. The Langmuir model agreed more with the adsorption process of CR and TC by WS<sub>400-4</sub>. Hence, the distribution of active sites on the adsorbent was relatively uniform, and the adsorption was uniform monolayer adsorption (Wang *et al.* 2019). At 323.15 K, the maximum adsorption capacity of CR and TC reached 116.41 and 58.58 mg·g<sup>-1</sup>, respectively. The negative values for  $\Delta G^\theta$  at different temperatures indicated that the adsorption of CR and TC occurred in a spontaneous process. In addition, as the temperature increased, a more significant decrease was observed in the  $\Delta G^\theta$  values, suggesting that higher temperature can be conducive to the CR and TC adsorption (Niu *et al.* 2021). The positive value of  $\Delta H^\theta$  indicated that the adsorption process was endothermic, and increasing the temperature was beneficial to the adsorption process. The value of  $\Delta S^\theta$  was positive, indicating that the entropy of the adsorption process increased, and the adsorption system occurred in the direction of increasing degrees of freedom at the solid-liquid interface (Maszkowska *et al.* 2014).

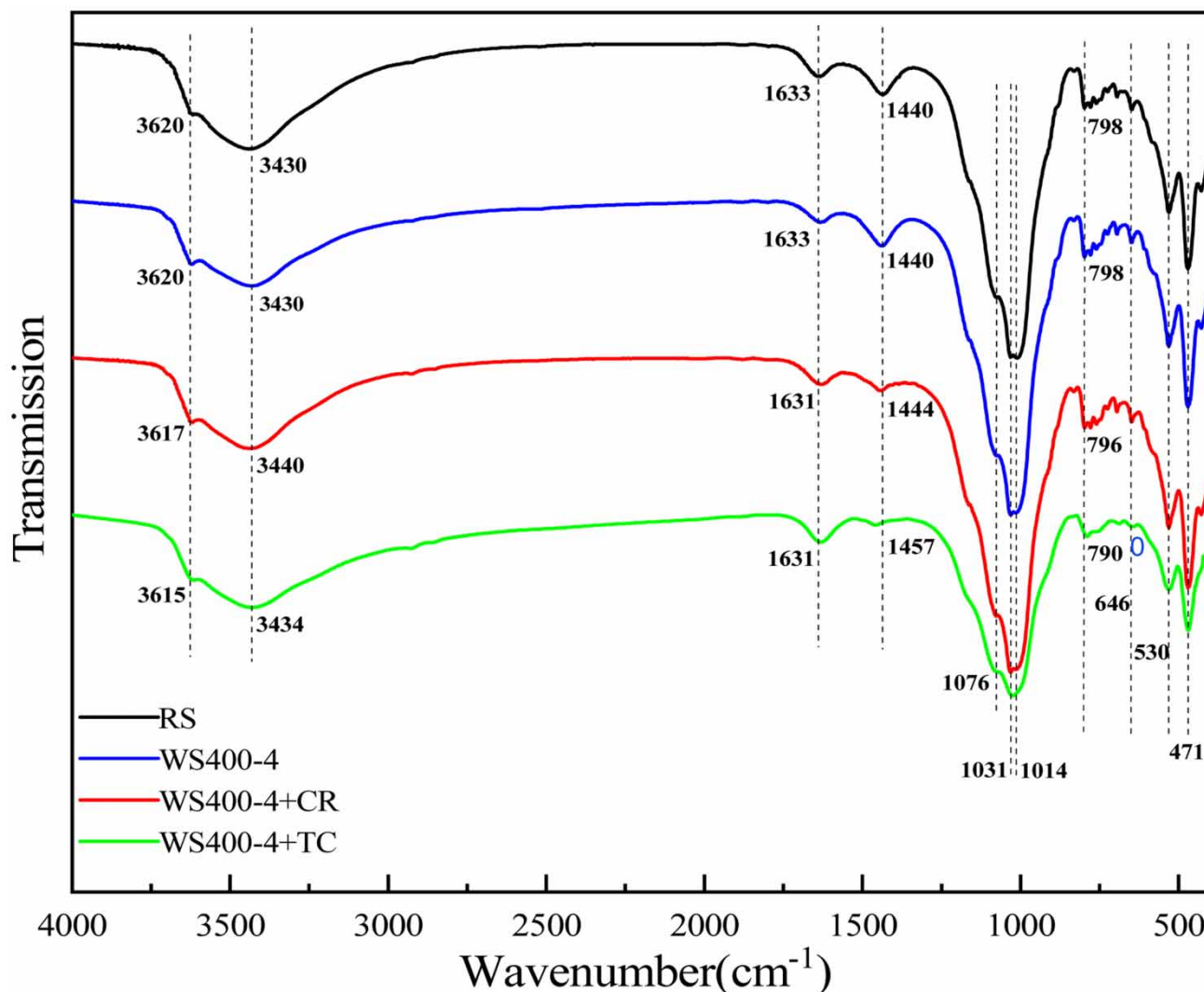
Table 4 lists the adsorption effects of different adsorbents on CR and TC in water. By comparing the adsorption effects in the following table, the  $q_{max}$  values of WS<sub>400-4</sub> in this study were 116.4 and 58.5 mg·g<sup>-1</sup>, respectively, which were higher than other biochars. Additionally, the WS<sub>400-4</sub> sample do not require modification, the preparation process of WS<sub>400-4</sub> is simple and efficient. Therefore, WS<sub>400-4</sub> exhibits great potential to become an excellent CR and TC adsorbent.

### 3.9. FTIR analysis

The FTIR spectra of RS, WS<sub>400-4</sub> and WS<sub>400-4</sub> after the adsorption of CR and TC are shown in Figure 9. There was no evident increase or decrease in the types of functional groups before and after thermal modification. The peaks at 3,620, 3,440 and 1,633 cm<sup>-1</sup> were the stretching and bending vibration peaks of O-H (Nibou *et al.* 2011). The peak at 1,440 cm<sup>-1</sup> was assigned to the C = C double bond of aromatic organic matter. The intensity of these peaks weakened in WS<sub>400-4</sub>, indicating the moisture and organic matter content in WTS after pyrolysis were significantly reduced. The Si-O stretching vibration peak near 1,002.8 cm<sup>-1</sup> in the intermediate frequency region, the characteristic absorption peak of Si-OH at 798 cm<sup>-1</sup>, the symmetrical stretching and asymmetric bending vibration peak of Si-O-Si at 746.3 cm<sup>-1</sup> and 580 cm<sup>-1</sup>, and Al-O stretching vibration peak near 476.3 cm<sup>-1</sup> basically did not change, indicating that the silicon oxide in WTS still maintained its stable structure after pyrolysis (Mezni *et al.* 2011; Yang *et al.* 2014). Some peak positions and heights shifted in the samples containing CR and TC

**Table 4** | Comparison between various adsorbents used for CR and TC adsorption

| Adsorbent raw material         | Contaminant | $q_{max}/\text{mg}\cdot\text{g}^{-1}$ | Reference                       |
|--------------------------------|-------------|---------------------------------------|---------------------------------|
| Red mud                        | CR          | 4.1                                   | Toor & Jin (2004)               |
| Graphene oxide/chitosan fibres | CR          | 294.0                                 | Du <i>et al.</i> (2014)         |
| Chitosan hydrobeads            | CR          | 92.6                                  | Chatterjee <i>et al.</i> (2007) |
| WS <sub>400-4</sub>            | CR          | 116.4                                 | This study                      |
| Kaolinite                      | TC          | 47                                    | Zhao <i>et al.</i> (2011)       |
| Vermiculite                    | TC          | 36.8                                  | Liu <i>et al.</i> (2017)        |
| Hydrothermal porous carbon     | TC          | 25.44                                 | Zhu <i>et al.</i> (2014)        |
| WS <sub>400-4</sub>            | TC          | 58.5                                  | This study                      |



**Figure 9** | FTIR spectra of RS and WS<sub>400-4</sub> before and after adsorption.

compared with the FTIR spectra of the WS<sub>400-4</sub> alone. After adsorption, the band at 3,620 cm<sup>-1</sup> in the WS<sub>400-4</sub> spectrum was shifted to 3,617 and 3,615 cm<sup>-1</sup>, the band at 3,430 cm<sup>-1</sup> spectrum was shifted to 3,440 and 3,434 cm<sup>-1</sup>, the band at 1,633 cm<sup>-1</sup> was shifted to 1,631 cm<sup>-1</sup>, and the band at 798 cm<sup>-1</sup> was shifted to 796 cm<sup>-1</sup> and 790 cm<sup>-1</sup>, suggesting that hydrogen bonds might form through the interactions of the hydroxyl groups on WS<sub>400-4</sub> and -COOH, -NH<sub>2</sub> and N-containing heterocyclic ring in the adsorbate (Akar *et al.* 2015). In addition, the delocalised  $\pi$  bond between TC heterocyclic ring and benzene ring has strong electron-absorbing property, as a  $\pi$  electron acceptor. There may be a conjugation effect between TC and WS<sub>400-4</sub> aromatised organic compounds (He *et al.* 2018). Therefore, after adsorption of TC onto WS<sub>400-4</sub>, the characteristic peak of the C = C double bond at 1,440 cm<sup>-1</sup> on the benzene ring significantly weakened and exhibited a blueshift, indicating that there was a certain  $\pi$ - $\pi$  stacking effect (Xiao *et al.* 2015). This phenomenon can also be observed in FTIR spectra after CR adsorption. In general, the efficient adsorption mechanism of WS<sub>400-4</sub> was mainly hydrogen bonding, accompanied by  $\pi$ - $\pi$  stacking.

#### 4. CONCLUSIONS

WS<sub>400-4</sub> was produced from WTS through O<sub>2</sub>-limited pyrolysis, and it displayed high specific surface area and well-developed porous structure. This low-cost adsorbent exhibited superior adsorption performance for both CR and TC. The adsorption process was more accurately described by the pseudo-second-order and Langmuir models. The entire process was jointly

controlled by liquid film diffusion, intra-particle diffusion and surface adsorption. CR and TC adsorption onto WS<sub>400-4</sub> were spontaneous, endothermic and randomness increase processes. According to the analysis, the adsorption process of CR was dominated by electrostatic attraction,  $\pi$ - $\pi$  conjugation and hydrogen bonding, whereas the adsorption process of TC was dominated by cation exchange, complex precipitation,  $\pi$ - $\pi$  conjugation and hydrogen bonding. WTS is a promising adsorbent with stable and efficient adsorption capacity, low cost, environmental protection and safety.

## DATA AVAILABILITY STATEMENT

All relevant data are included in the paper or its Supplementary Information.

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